

REINFORCEMENT LEARNING
SUBJECT CODE: 24DS742
GridWorld – Policy Evaluation and Value Iteration
Lab ASSIGNMENT - 3

Submitted by

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Objective

Formulate a GridWorld environment as a Reinforcement Learning problem, implement. Policy Evaluation and Value Iteration and compare different policies.

1. A 5x4 grid has been initialised in `WORLD_SPEC = np.array()`,
['*', '*', '*', '*'],
['*', ' ', ' ', '*'],
['*', ' ', ' ', '*'],
['*', 'g', '*', '*'],
['*', '*', '*', '*']

Where '*' represents a wall and the empty spaces represent a state from state space. The 'g' represents the goal state or terminal.

2. The states from the state space are $S = \{s1, s2, s3, s4, s5\}$, where `START_STATE = "S1"` and `GOAL_STATE = "S5"` or terminal.

The co-ordinates for the states are

"S1": (1, 1),

"S2": (1, 2),

"S3": (2, 1),

"S4": (2, 2),

"S5": (3, 1),

With walls ('*') assigned to the remaining co-ordinates from the grid.

3. Actions assigned to the states are "left", "right", "down", "up", where the actions space is `ACTIONS = ["<", ">", "v", "^"]` and `ACTION_EFFECTS = {"<": (0, -1), ">": (0, 1), "v": (1, 0), "^": (-1, 0),}`
4. Rewards being, `GOAL_REWARD = +1` and `OTHER_REWARDS = 0`. Example transitions from the executed program include S1 to S2 with reward 0, S1 to S3 with reward 0, and S3 to S5 with reward 1. Invalid movements (transition to wall / '*') leave the agent in the same state.
5. Additional parameters would be "gamma" being the discount factor = 0.9 and the convergence threshold (theta) = 1×10^{-6} .

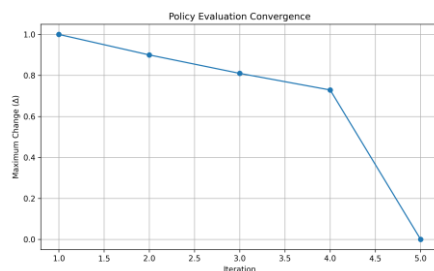
Task 1: Formulating the GridWorld Environment

RL Component	Description
States	$S = \{s1, s2, s3, s4, s5\}$
Actions	$ACTIONS = [<\", >\", \text{v}\", \text{^}\"]$
Reward Structure	+1 for g, 0 for remaining
Terminal State(s)	S5
Goal of the Agent	To find optimal policies.

Task 2: Policy Evaluation:

Fixed policy: S1 $\rightarrow$ Right, S2 $\rightarrow$ Down, S3 $\rightarrow$ Down, S4 $\rightarrow$ Left. The state-value function was iteratively updated until convergence.

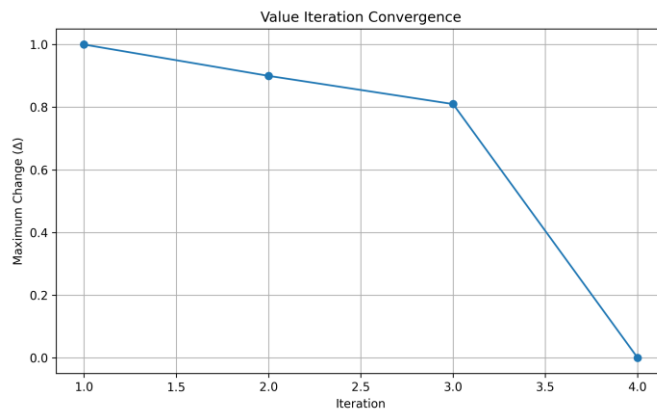
Iteration	Maximum Change (Δ)	Convergence Status
1	1.000000	Not Converged
2	0.900000	Not Converged
3	0.810000	Not Converged
4	0.729000	Not Converged
5	0.000000	Converged



Task 3: Value Iteration and Optimal Policy Generation:

Value Iteration considered all available actions at each state and selected the action with the maximum Bellman return. It converged after 4 iterations with final $\Delta = 0$.

State	Optimal Action	State Value
S1	Down ($\downarrow$)	0.900000
S2	Left ($\leftarrow$), Down ($\downarrow$)	0.810000
S3	Down ($\downarrow$)	1.000000
S4	Left ($\leftarrow$)	0.900000
S5	Terminal	0.000000



Task 4: Optimal Path Analysis:

The three policies were executed from S1 until S5 was reached. The random-policy path is the particular random trial produced by this execution.

Policy Type	Path Length	Total Reward	Goal Reached (Yes/No)
Random Policy	11	1.000	yes
Evaluated Policy	4	1.000	yes
Optimal Policy	2	1.000	yes

Random - S1 → S1 → S1 → S3 → S4 → S4 → S2 → S1 → S3 → S3 → S3 → S5

Evaluated - S1 → S2 → S4 → S3 → S5

Optimal - S1 → S3 → S5

The optimal policy reduced the path from 4 steps to 2 steps relative to the evaluated policy, a 50% reduction. Compared with the random trial, the 2-step path was 81.8% shorter. All successful runs received total reward 1.0 because the environment gives +1 at the goal and 0 on other transitions.

Analysis Questions:

What is a state in GridWorld?

The current position of the agent in the grid for each transition is called a state. In this 5 state environment, s1 to s5 are the states.

What actions can the agent perform?

The agent can perform these actions: “left”, “right”, “up”, “down”, if the requested action is not there, it remains in the same state.

What is the purpose of Policy Evaluation?

Policy Evaluation estimates the value of every state when the agent follows a fixed policy. The values are repeatedly updated using the Bellman expectation equation until convergence.

How does Value Iteration differ from Policy Evaluation?

Policy Evaluation evaluates a specified policy, whereas Value Iteration considers the available actions and selects the action with the highest expected return. Value Iteration therefore produces an optimal value function and corresponding policy.

What is the Bellman Optimality Equation?

$V^*(s) = \max_a [R(s,a) + \gamma V^*(s')]$. It states that the optimal value of a state is the maximum expected return over available actions. In this experiment, $\gamma = 0.9$.

Where, $R(s,a)$ is rewards, γ is the discount factor.

Which policy reached the goal using the fewest steps?

The Optimal Policy reached the goal in 2 steps, following $S1 \rightarrow S3 \rightarrow S5$. The Evaluated Policy required 4 steps and the observed Random Policy required 11 steps.

Why is the optimal policy superior to a random policy?

The optimal policy selects actions using estimated future returns, while a random policy does not account for the consequences of its actions. In this experiment, the optimal policy reached the goal in 2 steps compared with 11 steps for the observed random trial.

How would obstacles affect the optimal path?

Obstacles would restrict available transitions and could make the direct route impossible. The optimal policy would then choose an alternative route around the obstacles, potentially increasing the number of steps and changing optimal actions.

Conclusion:

The 5-state GridWorld experiment demonstrated the roles of Policy Evaluation and Value Iteration in reinforcement learning. Policy Evaluation estimated state values for a fixed policy and converged after five iterations, while Value Iteration determined the optimal values and

actions and converged after four iterations. The observed random, evaluated, and optimal policies required 11, 4, and 2 steps respectively to reach the goal. Although all three successful runs obtained a total reward of 1.0 under this reward structure, the optimal policy achieved the goal most efficiently. Obstacles would modify the transition structure and could force the optimal policy to use longer alternative routes.